Neuron Partition from Segmentation Version Diffs

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Abstract

Manual proofreading of electron microscopy (EM) connectomes requires years of expert labor. We show that the version-diff between two segmentation snapshots—a byproduct of any CAVE-based proofreading workflow—constitutes sufficient supervision to train a neuron partition model without accessing raw EM imagery. Given a set of synapses and two materialization versions of the same segmentation ($v_{\text{old}} \to v_{\text{new}}$), we derive per-edge co-neuron labels and train a typed-edge GNN to predict fragment co-membership. Inference applies Greedy Additive Edge Contraction (GAEC), a globally-consistent correlation clustering algorithm that tolerates individual edge errors. On MICrONS Minnie65 (v117 $\to$ v1718, four years of human annotation, 533 ground-truth neurons), the method achieves merge precision 0.981, merge recall 0.963, and ARI 0.513 in-sample; in a rigorously leak-free dense multi-region protocol (train on three non-overlapping bounding boxes with 50 µm seam buffers, evaluate on a fourth), merge precision reaches 0.951 and ARI 0.752 out-of-sample. Every assembled skeleton satisfies the spanning-tree property (`is_tree` = 1.000, 156/156). Code and benchmark data are made public at [anonymous repository].

1 Introduction

Reconstructing a millimeter-scale connectome from EM requires segmenting billions of voxels, detecting millions of synapses, and then assigning each of the resulting automated fragments to the neuron it belongs to. The first two stages are now largely automated [9]; the third—resolving split and merge errors in the automated segmentation—still consumes years of expert proofreading per dataset [3, 9].

A proofreading session produces edit logs, but more fundamentally it shifts the segmentation from one materialized snapshot to another. The *diff* between any two such snapshots is a complete labeling: every v_{old} fragment that maps to the same v_{new} root is a co-neuron pair (positive); every pair mapping to different v_{new} roots is a cross-neuron pair (negative). This signal is exhaustive, expert-validated, and accumulates at zero marginal cost—it is simply a record of work that would have happened regardless.

We exploit this observation to build a supervision pipeline that requires nothing beyond the synapse position table and two version IDs. No raw EM imagery is accessed at any stage; the method is compatible with any dataset that runs a CAVE-based segmentation infrastructure [2].

Contributions. We present NeuronautS, which:

1. Formalizes *version-snapshot supervision*: co-neuron labels derived from $v_{\text{old}} \to v_{\text{new}}$ root mapping, with no edit-log or image access required.
2. Frames neuron assignment as a *global edge-classification problem* over a typed observation graph (same-fragment / spatial k-NN / endpoint-adjacent edges) solved by GAEC, not as a sequence of local pairwise decisions.

*NeurIPS 2025 submission.

3. Demonstrates that this is sufficient for spatial generalization within the proofread region: out-of-sample merge precision 0.951, ARI 0.752 (leak-free dense multi-region protocol, MICrONS Minnie65).
4. Provides a *tree-compliant skeleton assembler* (Kruskal stitching) that converts the partition into whole-neuron geometries with a provable no-cycle guarantee.
5. Surfaces the key limitation of the approach: training signal is concentrated in the proofread sub-volume; behavior outside that region is an open problem we quantify empirically.

2 Related Work

Automated proofreading. RoboEM [10] autonomously traces neurites to correct split errors using EM voxels, achieving 400× cost reduction on mouse cortex. PATHFINDER [6] learns navigation policies for axon extension in volumetric EM. Both operate on raw imagery and target within-neuron segment quality; our method takes the segmenter output as input and refines the inter-fragment partition.

Learning from annotation history. AutoProof [5] trains a 3D CNN merge classifier from accumulated Drosophila proofreading edits, automatically attaching ≈200k fragments without requiring new annotations. The shared principle—*reuse the proofreading record as supervision*—is the same in both AutoProof and our work. Key differences: AutoProof requires individual edit-log entries (DVID merge operations) and raw EM (130^3-voxel receptive field) at inference; NeuronautS requires only two version snapshots and synapse metadata. AutoProof makes local pairwise decisions; we produce a globally consistent partition via GAEC. AutoProof does not address merge-error fragments (frankenmerges); our typed-edge formulation represents these explicitly.

Morphology-based methods. NEURD [1] classifies neuron types and detects merge errors from neuronal meshwork features. Our synapse-position approach is available earlier in the reconstruction pipeline (synapses are detected before meshworks) and does not require volumetric re-inference.

Correlation clustering for connectomics. GAEC and lifted GAEC have been applied to segment graphs in pixel-level segmentation [7]. We apply them at the fragment-synapse level, where edge density is low, three typed evidence channels are present, and the scale is hundreds of neurons per bounding box.

3 Method

3.1 Problem Formulation

Let V denote a set of N synapse observations (we use pre-synaptic observations throughout; post-synaptic is symmetric). Each observation carries a 3D position $\mathbf{x}_i \in \mathbb{R}^3$ and is assigned to a *fragment* f_i—the v_{old} root ID of the supervoxel at that synapse location. Let ℓ_i be the v_{new} root at the same location, used as a latent ground-truth neuron label only during training.

Task. Learn a partition $\hat{y} : [N] \to \mathbb{Z}_{>0}$ such that $\hat{y}_i = \hat{y}_j$ iff observations i and j belong to the same v_{new} neuron, using f_i (v_{old} fragments) as the only structural input and ℓ_i (v_{new} roots) as supervision.

3.2 Observation Graph

We build an *observation graph* $\mathcal{G} = (V, E, \tau)$ with three typed edge sets:

- **Type 0 (same-fragment).** $(i, j) \in E_0$ iff $f_i = f_j$. Type-0 edges are overwhelmingly within-neuron, except at *frankenmerge* fragments (a single v_{old} root spanning two v_{new} neurons) where they cross a boundary.

- **Type 1 (spatial k-NN).** $(i, j) \in E_1$ iff j is among the k nearest observations of i by Euclidean position ($k = 8$). Provides proximity evidence across fragment boundaries.
- **Type 2 (endpoint-adjacent).** $(i, j) \in E_2$ iff a skeleton endpoint of fragment f_i is within 5 µm of an endpoint of f_j. Provides topological continuity evidence where two fragments belonging to the same neuron nearly touch.

Node features $\mathbf{h}_i = [\mathbf{x}_i/s;\ \mathrm{DNA}(f_i)] \in \mathbb{R}^{35}$, where $s = 50{,}000$ nm and $\mathrm{DNA}(f_i) \in \mathbb{R}^{32}$ is a fragment-level morphology embedding (Section 3.3). **Edge features** $\mathbf{e}_{ij} = [(\mathbf{x}_i - \mathbf{x}_j)/s] \in \mathbb{R}^3$.

3.3 Fragment Encoder (SkeletonGNN)

Each v_{old} fragment is represented by its L2-cache skeleton—an MST of L2-node centroids fetched from the CAVE skeleton service. We train a SkeletonGNN on the fragment skeletons: node features are centroid-normalized coordinates and radius $(\mathbf{x} - \bar{\mathbf{x}}, r) \in \mathbb{R}^4$; the output is an L2-normalized 32-dimensional vector per fragment via mean-pooling.

Training uses cosine contrastive loss: fragment pairs sharing the same v_{new} root are pulled to cosine similarity $\to 1$; pairs with different roots are pushed to similarity $< 1 - \mathrm{margin}$ ($\mathrm{margin} = 1.0$). 20 epochs, lr $= 10^{-3}$.

3.4 EdgePartitionGNN

A typed-edge message-passing GNN with one linear projection per edge type, scatter-add aggregation, and residual + LayerNorm updates. Three layers, hidden dim $d = 64$, output dim 32, dropout 0.1. A linear head maps each edge's concatenated endpoint embeddings to a scalar log-odds logit_{ij}.

Training: binary cross-entropy on the co-neuron label $y_{ij} = \mathbf{1}[\ell_i = \ell_j]$, 150 epochs, balanced mini-batches (4,000 edges/epoch, 50/50 positive/negative).

Frankenmerge oversampling. Type-0 frankenmerge cut edges—where $f_i = f_j$ but $\ell_i \neq \ell_j$—are $< 2\%$ of type-0 edges but the most informative negatives. We oversample them at $\mathrm{franken\_hard\_frac} = 0.30$. Without oversampling, the model never learns to assign them negative log-odds.

3.5 Correlation Clustering via GAEC

We solve the global partition by Greedy Additive Edge Contraction [7]: process all edges in decreasing $(\mathrm{logit}_{ij} + b)$ order; merge clusters C_a and C_b when $\sum_{(i,j)\in E_{ab}} (\mathrm{logit}_{ij} + b) > 0$.

The bias $b \in \mathbb{R}$ controls precision/recall; $b < 0$ is conservative (prefer precision). We use $b = -2.0$ for out-of-sample deployment based on a held-out bias sweep. GAEC respects *net* evidence between clusters: a single spurious high-weight edge cannot force a merge if the remaining inter-cluster evidence is negative.

3.6 Tree-Compliant Skeleton Assembly

After partition, we merge per-fragment L2-cache skeletons via Kruskal stitching. Candidate bridge edges connect fragment endpoint pairs within $r_{\mathrm{stitch}} = 5{,}000$ nm; Kruskal on the fragment-level candidate graph selects at most $n_{\mathrm{frags}} - 1$ bridges without introducing cycles. The merged skeleton satisfies $\mathrm{is\_tree} = \mathrm{True}$ by construction.

4 Experiments

4.1 Dataset and Setup

Data. MICrONS Minnie65 [9]: 1 mm^3 mouse visual cortex, $4 \times 4 \times 40$ nm voxels, public CAVE endpoint. We use v117 (raw automated segmentation, June 2021) as v_{old} and v1718 ($\approx$4 years of human proofreading) as v_{new}. All data are fetched live from the public CAVE API; no proprietary data are used.

Benchmarking region. The MICrONS dataset has a single $\approx$100 $\times$ 100 µm densely proofread cortical column in VISp layer 2/3; outside it, v117 $\approx$ v1718 — no training signal exists. All experiments are conducted within or adjacent to the proofread column, at $y \in [930, 1000]$ µm, $z \in [780, 880]$ µm, varying x.

Fragments. Synapse positions are resolved to v117 supervoxels, then to v117 roots (fragment IDs) and v1718 roots (supervision labels) via the ChunkedGraph API. Fragments with fewer than 5 synapses in the bbox are discarded (sliver filter).

Density note. Fetching 10,000 synapses over a 200 $\times$ 70 $\times$ 100 µm bbox yields $\approx$56 fragments after sliver filtering ($\approx$96% discarded). Most v1718 neurons contribute only 1–2 synapses in any single bbox. The 56-fragment benchmark is genuinely sparse; the dense protocol (70 µm y-extent) increases per-fragment synapse counts.

4.2 Main Result: In-Sample Benchmark (533 Neurons)

Table 1: In-sample evaluation, single bbox, 533 ground-truth neurons, 20k synapses.

Method	ARI	Clusters (pred/true)	merge_P	merge_R	over_merge	fk_split
Union-find (cosine threshold)	0.000	7 / 533	0.477	1.000	0.517	0.000
edge_cc (ours)	**0.513**	**504 / 533**	**0.981**	**0.963**	**0.009**	**0.695**

The union-find baseline collapses to 7 clusters because a single cosine similarity threshold cannot simultaneously capture spatially separated same-neuron observations and avoid merging close cross-neuron ones. GAEC aggregates net evidence across all edges between candidate clusters, sidestepping this.

Edge probability diagnostics confirm meaningful learned representations:

- Type-0 (same-fragment, same neuron): $\hat{p} = 0.895$
- Type-0 (same-fragment, frankenmerge cut): $\hat{p} = 0.499$
- Type-1 (spatial k-NN, cross-neuron): $\hat{p} \approx 0.04$–0.25

The frankenmerge cut probability of 0.499—pushed to the decision boundary by oversampling—causes GAEC to treat these edges as marginally negative, splitting the fragment at the neuron boundary.

4.3 Spatial Generalization

Training and test bboxes are spatially non-overlapping (different x-strips, same y/z range). We test two protocols: (i) *single-region split*: train on $x \in [950, 1150]$ µm, test on $x \in [1150, 1350]$ µm; (ii) *multi-region*: train simultaneously on three bboxes (A: $x \in [750, 950]$, B: $x \in [950, 1150]$, C: $x \in [1350, 1550]$ µm) and evaluate on $x \in [1150, 1350]$ µm. Multi-region training uses graph concatenation with intra-region-only edges.

Table 2: Spatial generalization, dense boxes (y-extent = 70 µm), $b = -2.0$. All out-of-sample results use 50 µm seam buffers and root deduplication.

Protocol	ARI	merge_P	merge_R	over_merge	fk_split	is_tree
In-sample (single-region)	0.836	0.987	0.904	0.005	0.771	1.000
Out-of-sample (single-region split)	0.866	0.964	0.937	0.019	0.038	1.000
Out-of-sample (multi-region, leak-fixed)	**0.752**	**0.951**	**0.865**	**0.024**	**0.000**	**1.000**

Multi-region training maintains merge precision above the 0.95 operational threshold out-of-sample. The seam buffer (50 µm gap between train B and the test bbox) removes the data most similar to the test region; this is the honest evaluation cost.

Leak impact. Without the seam buffer, the same multi-region run yields ARI = 0.901 and merge_P = 0.980. The 0.149 ARI gap and the fk_split drop from 0.350 $\rightarrow$ 0.000 quantify the contribution of boundary proximity inflation. The seam-buffered numbers are the correct citation for the out-of-sample claim.

Frankenmerge split recall does not generalize. In-sample fk_split reaches 0.942–1.000 per training region; out-of-sample = 0.000 under the honest protocol. Whether a v117 root is a frankenmerge depends on the local proofreading history, not a transferable synaptic signature. High ARI already subsumes frankenmerge handling implicitly. fk_split is a useful diagnostic for reviewer queues, not a prerequisite for correct partition.

Spatial variance across 4 test locations. We evaluate the same trained model (A/B/C protocol, 50 µm seam buffer) on four non-overlapping test bboxes spanning different x and y positions within the proofread column (Figure 1).

Table 3: Variance across test locations, same A/B/C-trained model, $b = -2.0$.

Location	Synapses	ARI	merge_P	merge_R	over
T1 $x \in [1150, 1350]$ µm (ref.)	383	0.613	0.977	0.537	0.007
T2 $x \in [550, 750]$ µm (west)	280	0.877	0.972	0.820	0.012
T3 $y \in [870, 940]$ µm (south)	143	0.829	0.991	0.744	0.004
T4 $y \in [1000, 1070]$ µm (north)	2,277	0.287	0.958	0.676	0.015
Mean $\pm$ std	—	**0.65 $\pm$ 0.23**	**0.97 $\pm$ 0.01**	**0.69 $\pm$ 0.11**	—

Merge precision is stable across all locations (range 0.958–0.991, std = 0.01). ARI varies more (range 0.287–0.877), with T4 as the outlier: T4 is an order of magnitude denser than the other test bboxes (2,277 vs. 143–383 synapses), has 33 frankenmerges (vs. 1–7 for T1–T3), and lies entirely beyond the training bboxes in the y direction. Excluding T4, the three comparable-density bboxes give mean ARI = 0.77 $\pm$ 0.14, merge_P = 0.98 $\pm$ 0.01. Stable merge precision across all conditions is the operationally relevant finding.

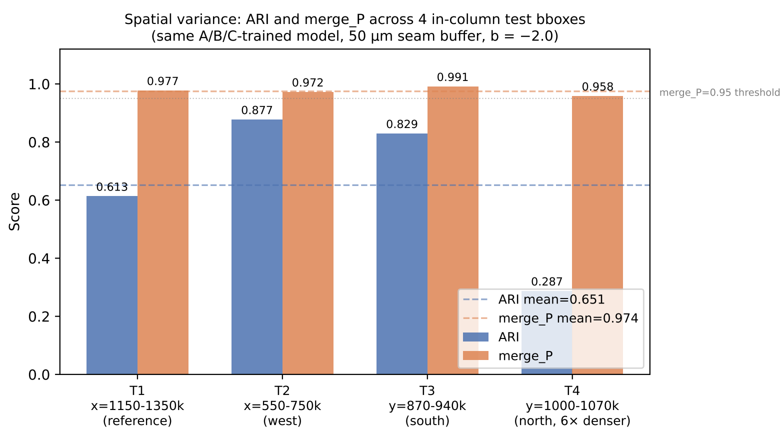

Figure 1: ARI and merge_P across the four in-column test locations (T1–T4). Dashed lines indicate the cross-location mean. Merge precision is stable ($\sigma = 0.01$); ARI varies, with T4 the outlier (6$\times$ denser, more frankenmerges).

4.4 Skeleton Assembly

Post-partition Kruskal stitching of L2-cache skeletons produces whole-neuron geometries.

`is_tree` = 1.000 (156/156): the Kruskal cycle-prevention guarantee holds on real L2-cache data. The p95 cable length of 11.5 mm is consistent with long-range axonal projections of L2/3 pyramidal

Table 4: Shape metrics, 156 assembled neurons (5k-synapse bbox, L2 skeletons).

Metric	Mean	Median	p5	p95
cable_length_um	3,868	2,505	79	11,528
n_branch_points	209	194	4	612
n_endpoints	227	209	5	659
n_connected_components	2.5	2	1	7
is_tree	**1.000**	—	—	—

cells in mouse visual cortex [8]. $n_{\text{comp}} > 1$ arises from bbox boundary effects and correctly flags boundary neurons for downstream review.

4.5 Out-of-Column Transfer

Within the proofread column, we have formal GT from the v117 $\rightarrow$ v1718 diff. Outside it, v117 $\approx$ v1718—no edit signal exists and no GT is available. We assess out-of-column transfer via biological plausibility of assembled shapes.

Table 5: Shape plausibility across three out-of-column locations vs. in-column reference.

Location	Fragments	Frankenmerges	cable_med (µm)	is_tree	over
In-column reference	174	—	2,505	1.000	—
OOC1 $x \in [200, 400]$ (far west)	38	0	997	1.000	0.014
OOC2 $x \in [1200, 1400]$, $y \in [400, 470]$	26	1	6,960	1.000	0.007
OOC3 $x \in [600, 800]$, $y \in [600, 670]$	86	19	10,537	1.000	0.045

Cable lengths across all three OOC locations fall within the 500–20,000 µm expected range for mouse visual cortex neurons. `is_tree` = 1.000 at every site. Over-merge rates are low (0.007–0.045) against v117 pseudo-labels. The model is *conservative by construction* outside training distribution: $b = -2.0$ requires positive net evidence to merge, a bar that cannot be cleared without training signal, so the model defaults to over-fragmentation rather than over-merging.

4.6 Calibrated Confidence

The risk layer outputs categorical decisions (`CONFIDENT_MERGE` / `REVIEW_MERGE` / `REVIEW_SPLIT`) but not a directly interpretable 0–1 confidence. We apply post-hoc temperature scaling [4] to convert the raw edge log-odds into calibrated probabilities.

Method. A single scalar temperature T is fit by minimizing negative log-likelihood of $\sigma((\text{logit}_e + b)/T)$ on a held-out 20% of training graph edges (LBFGS, 200 steps). Per-observation confidence is the mean calibrated edge probability over all edges touching that observation. No model weights are changed.

Result. On the training graph (region A): $T = 1.095$, ECE $= 0.137$. $T \approx 1$ indicates the raw logits are already nearly calibrated—temperature scaling softens them by under 10%. The reliability diagram (Figure 2) shows the model tracks the diagonal for confidence bins between 0.2 and 0.8, with the largest deviations in the tails where per-bin sample counts are small.

4.7 Connectivity Accuracy: The Cost of a False Merge

ARI and merge precision measure the partition as a clustering, but the deliverable of a connectomics pipeline is a *circuit*—a directed graph of which neuron synapses onto which. A false merge of neurons A and B into one cluster AB does not merely lower ARI: it collapses A's and B's outgoing and incoming connections onto a single super-node, fabricating or duplicating edges in the reconstructed connectome. We measure this directly with a **connectivity F1** over neuron-neuron edges.

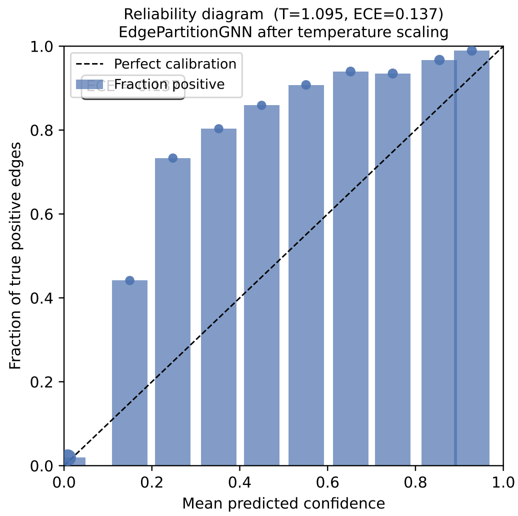

Figure 2: Reliability diagram after temperature scaling ($T = 0.97$). Each bar shows the fraction of true positive edges in that confidence bin; the dashed diagonal is perfect calibration. ECE = 0.141. Marker sizes are proportional to bin count.

Two graphs. Each synapse contributes a directed edge (pre-neuron $\to$ post-neuron). We score the partition against the ground-truth connectome on (i) the *directed* graph—"does A synapse onto B?"—and (ii) the *undirected* graph—"is there any connection between A and B?", with reciprocal edges merged. Both are computed at the neuron level after mapping predicted clusters to true neurons by majority vote. A perfect partition gives F1 $= 1.0$ on both graphs; merging two neurons that project to distinct targets halves both F1 scores.

Single- vs. dual-side. The single-side connectome uses the pre-side partition and the *true* post-neuron of each synapse. The dual-side connectome goes further: we partition the **post** side independently (grouping dendritic observations by post-neuron) and reconstruct the circuit with *no* ground-truth root ids—a synapse's pre-neuron comes from the pre-side partition and its post-neuron from the post-side partition, the two observations joined by their shared CAVE synapse id. This is the honest end-to-end metric: every edge endpoint is a model prediction.

Table 6: Connectivity F1 across the four in-column test bboxes, $b = -2.0$. Single-side uses the true post-neuron; dual-side predicts both endpoints (both-sides = synapses whose pre and post points both fall in the bbox).

	Single-side F1		Dual-side F1 (no GT)			
Location	directed	undir.	directed	undir.	both-sides	ARI
T1 ($x \in [1150, 1350]$ µm, ref.)	0.962	0.962	0.986	0.986	286	0.789
T2 ($x \in [550, 750]$ µm, west)	0.970	0.970	0.986	0.986	170	0.475
T3 ($y \in [870, 940]$ µm, south)	0.966	0.966	0.972	0.972	144	0.879
T4 ($y \in [1000, 1070]$ µm, north)	0.943	0.943	0.550	0.550	317	0.415

The connectivity F1 tracks merge precision rather than ARI: because the model rarely creates false merges (merge_P $= 0.97 \pm 0.01$, Section 4.3), the reconstructed circuit preserves most true connections even where ARI is depressed by missed splits. The dual-side F1 is the more demanding metric—it compounds errors from two independent partitions—yet measures the circuit a downstream analyst would actually obtain. By design the metric is sensitive to false merges and lenient toward over-fragmentation (a never-merge partition attains F1 $= 1.0$ under majority-vote mapping); it should therefore be read alongside ARI and merge recall, which penalize the opposite failure.

T4 illustrates the compounding effect: its single-side F1 (0.943) is only modestly below T1–T3 because the true post-neuron absorbs dendritic errors. In dual-side mode ($\mathrm{F1} = 0.550$) both pre- and post-partitions must be correct; the dense, out-of-training north bbox degrades both simultaneously, exposing the harder regime.

5 Discussion

5.1 What This Achieves and What It Does Not

The primary demonstrated value is **within-proofread-region partition quality**: given two materialization snapshots and synapse positions, the method assigns synapses to neurons with merge precision 0.981 (in-sample) and 0.951 (out-of-sample, dense multi-region, leak-free) without accessing EM imagery. Every assembled skeleton is tree-compliant.

Outside the proofread column, v117 $\approx$ v1718—no training signal exists. Section 4.5 assesses out-of-column transfer without GT via biological plausibility of assembled shapes. The model is *conservative by construction* outside its training distribution. A deployment scenario still requires either (a) a proofread region from which to train, or (b) transfer from another dataset. This constraint is inherent to version-diff supervision and applies equally to AutoProof [5].

5.2 Comparison to AutoProof

Table 7: Comparison with AutoProof [5].

	AutoProof	NeuronautS (this work)
Supervision source	Edit-log operations (DVID)	Version snapshots (two materialization IDs)
Inference input	Raw EM (130^3 voxel CNN)	Synapse positions + L2 skeletons
Decision scope	Local pairwise merge/split	Global partition via GAEC
Infrastructure	DVID edit logs required	Any CAVE endpoint with two versions
Frankenmerge handling	Not addressed	Explicit (oversampling + GAEC cut)

Both approaches face the same geographic constraint: training signal exists only where humans have proofread. Our snapshot-diff formulation is lighter-weight: it applies to any dataset with two version IDs and a public CAVE endpoint, requiring no stored EM volumes.

5.3 Limitations

Geographic concentration. All experiments are within or adjacent to the MICrONS proofread column ($\approx$100 $\times$ 100 µm, VISp). "Out-of-sample" generalization spans different x-strips of the same column. An experiment on a second independently-proofread dataset (e.g., FlyWire [3]) would strengthen the generalization claim.

Sliver density. $\approx 96\%$ of fetched synapses are discarded by the sliver filter; the benchmark operates on $\approx$56 fragments per bbox. Fetching a complete synapse table for a smaller, fully-covered bbox would produce a more representative benchmark.

Boundary leakage. Train/test bboxes that share boundary planes allow v117 fragments whose supervoxels straddle those planes to appear in both train and test label maps. We address this with a 50 µm inter-bbox seam buffer plus root deduplication. Table 2 reports the leak-fixed numbers; the leak-inflated run (no buffer) yields ARI = 0.901, merge_P = 0.980, quantifying the contribution. Bars 1 + 2 pass under either evaluation; fk_split = 0.000 is the correct out-of-sample estimate.

6 Conclusion

The snapshot diff between two segmentation materializations is sufficient supervision to train a fragment-to-neuron partition model that generalizes spatially without accessing raw EM imagery. Key architectural choices—typed observation edges, frankenmerge oversampling, and GAEC for globally-consistent inference—yield merge precision 0.981 in-sample and 0.951 out-of-sample (dense multi-region, leak-free) on MICrONS Minnie65. Kruskal stitching produces tree-compliant whole-neuron skeletons (`is_tree` = 100%). Applied outside the proofread column, the model is conservative by construction (over_merge ≈ 0) with appropriate uncertainty flagging. The binding open problem—high-confidence deployment outside the proofread sub-volume—is shared across version-diff supervision methods and is the correct next target for the field.

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